Nereus-LC: Event-Triggered Dog Waste Detection and Response

Jesse Beneventi
Washington State University
Student ID: 011848749
jesse.beneventi@wsu.edu

Abstract

Uncontrolled pet activity on residential lawns can cause localized turf damage, odor, and sanitation concerns, while conventional timer-based irrigation systems cannot detect or respond to these events. This project presents NEREUS-LC, an event-triggered computer vision prototype for detecting dog activity and logging likely waste events. The current system combines YOLOv8 dog detection, a fine-tuned MobileNetV2 posture gate, CLIP-based pee/poop/neutral confirmation, exponential moving average (EMA) smoothing, and hysteresis-based event logic. When a likely event is detected, the system saves clean reference snapshots, post-event snapshots, localized region crops, and optional annotated video for review. A MobileNetV2 poop-posture classifier achieved 0.995 test accuracy and an F1 score of 0.994 at $\tau^\star = 0.53$, while CLIP experiments showed that prompt wording strongly affects pee/poop/neutral scores and can introduce a bias toward the pee label. The current prototype demonstrates a hybrid runtime test harness rather than a finished autonomous cleanup system. Remaining work includes expanding hard-negative examples, calibrating event thresholds, improving pee/poop separation, and connecting the vision pipeline to a physical response.

Figure 1: Concept image of the proposed Nereus-LC yard post installed at the edge of a lawn. The design combines an integrated camera with a directional sprinkler head to support event-triggered monitoring and localized response without treating the entire yard.

1 Introduction

Problem significance. Pet ownership is widespread—roughly 66–71% of U.S. households include at least one animal [Insurance Information Institute, 2025, Forbes Advisor, 2025]. While pets enrich daily life, their unsupervised activity creates recurring lawn-maintenance challenges. Defecation and urination can contribute to localized odor, sanitation concerns, nitrogen overload, and turf discoloration [PetMD Editors, 2025]. Timer-based irrigation cannot detect or respond to these localized events, leading to water waste and missed opportunities for targeted remediation.

Motivation. Recent advances in low-cost RGB cameras, object detection, lightweight neural networks, and vision-language models make real-time outdoor perception increasingly practical. Instead of watering or treating an entire lawn on a fixed schedule, an event-triggered system can focus on detecting when a relevant animal activity occurs, saving evidence from that event, and supporting a targeted response.

Project goals and contributions. This project develops NEREUS-LC, a computer vision prototype for event-triggered dog waste detection and response. The current implementation includes: (i) YOLOv8 dog detection, (ii) a MobileNetV2 posture gate for likely poop/squat events, (iii) CLIP-based pee/poop/neutral confirmation on event crops, (iv) EMA smoothing and hysteresis to stabilize event decisions, and (v) snapshot, crop, and annotated-video logging for later review. Earlier peeing-classifier experiments also helped identify the need for more diverse positives, hard negatives, and better threshold calibration.

Existing methods overview. Commercial deterrent and irrigation products generally provide fixed responses but lack perception. Computer vision tools such as YOLOv8 [Jocher and Ultralytics, 2023], MobileNetV2 [Sandler et al., 2018], and CLIP [Radford et al., 2021] provide relevant building blocks for detecting animals, classifying posture, and comparing image crops against text descriptions. However, the main challenge in this project is not only classification accuracy on still images, but stable event-level behavior in video.

Plan overview. Figure 2 summarizes the updated NEREUS-LC pipeline and separates completed components from features that still require additional work. The current prototype processes RGB video with YOLOv8 dog detection, generates candidate event crops, applies CLIP-based pee/poop/neutral confirmation, smooths event scores over time, and saves event logs, snapshots, localized crops, and annotated video. MobileNetV2 posture gating remains part of the prototype but still needs refinement in the full runtime setting, especially for separating pee-related behavior from poop posture and neutral dog activity. Ultrasonic deterrence and physical response integration are planned next steps, while ExG/VARI lawn-health analysis remains a future extension that has not yet been implemented.

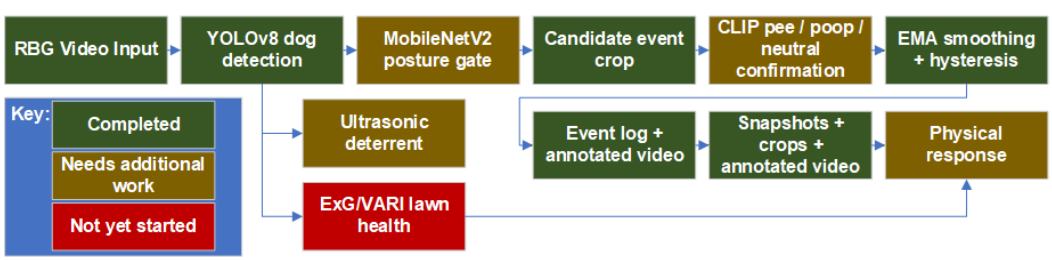


Figure 2: Updated Nereus-LC pipeline showing completed, in-progress, and future components. The current prototype includes RGB video input, YOLOv8 dog detection, candidate event cropping, CLIP confirmation, EMA/hysteresis stabilization, event logging, snapshots, crops, and annotated video. Deterrence, physical response, and ExG/VARI lawn-health analysis remain future extensions.

2 Literature Review

2.1 Computer Vision Foundations

Real-time object detection. One-stage detectors such as YOLOv8 balance accuracy and speed for real-time perception tasks [Jocher and Ultralytics, 2023]. In this project, YOLOv8 is used as an upstream dog detector rather than as the full waste-event classifier. This allows the system to first localize the animal, then pass a smaller crop to downstream posture and event-confirmation modules.

Lightweight posture classification. MobileNetV2 provides an efficient convolutional backbone for image classification on limited datasets and edge-oriented systems [Sandler et al., 2018]. Its low computational cost makes it useful as a posture gate in the runtime pipeline. Rather than treating MobileNetV2 as the final event detector, this project uses it to identify likely squat/posture frames that should receive additional confirmation.

Vision-language confirmation. CLIP learns a shared representation between images and natural-language descriptions, allowing an image crop to be scored against prompts such as "a dog peeing on grass," "a dog pooping on grass," or "a dog standing on grass" [Radford et al., 2021]. In this project, CLIP is used as a lightweight confirmation layer for pee/poop/neutral evidence. The early results show that prompt wording matters: poorly calibrated prompts can cause pee/poop confusion, so event thresholds and hard-negative examples remain important.

Animal behavior and waste-event detection. Formal literature on dog feces and urination detection is limited, but related behavior-sensing systems such as DeePosit show interest in detecting animal waste events using computer vision [Martinez et al., 2024]. This project contributes a practical event-triggered pipeline that combines dog detection, posture gating, CLIP confirmation, temporal smoothing, and post-event evidence capture.

Summary and gap. Existing work provides strong components for detection, classification, and image-text matching, but these tools do not directly solve the full event-level problem. Nereus-LC focuses on the gap between frame-level model predictions and stable runtime behavior: detecting a dog, deciding whether an event is likely, confirming the event type, and saving useful evidence for future response integration.

3 Technical Approach

3.1 System Overview

The current version of NEREUS-LC implements a hybrid runtime computer vision pipeline for detecting dog activity, identifying likely waste-event behavior, and saving evidence for review. Video input comes from either a live RGB camera or a prerecorded MP4 file. YOLOv8 is used as the upstream detector to locate dogs in each frame. A fine-tuned MobileNetV2 classifier then acts as a posture gate for likely poop/squat events. When the posture gate indicates that an event may be occurring, the system evaluates a localized crop with CLIP to estimate whether the event is more consistent with pee, poop, or neutral dog behavior.

The *NereusWatch* runtime handles video input, dog detection, posture scoring, CLIP confirmation, probability smoothing, ON/OFF thresholding, target-region cropping, clean snapshot capture, post-event snapshot capture, optional annotated video recording, and console logging. The current system is therefore best understood as a hybrid event-detection test harness rather than a finished autonomous cleanup product. Its purpose is to test whether multiple lightweight perception components can work together to produce stable event-level decisions from noisy outdoor video.

The architecture is guided by two constraints: (i) stable event detection under real-world outdoor variability and (ii) practical integration with a future physical response. YOLOv8 provides dog localization, MobileNetV2 provides fast posture gating, and CLIP provides a flexible image-text confirmation layer. EMA smoothing and ON/OFF thresholding are then used to convert noisy frame-level scores into more stable event-level behavior.

3.2 Dog Detection and Posture Gating

YOLOv8 dog detection. YOLOv8n is used as the first stage of the pipeline. Each frame is passed through the detector, and only detections labeled as dogs are retained. The largest dog detection is selected for downstream processing. This detection stage reduces the search area for later modules and allows the rest of the system to focus on a dog-centered crop rather than the entire frame.

MobileNetV2 posture gate. A MobileNetV2 backbone pretrained on ImageNet is used for poop-posture recognition. The original 1000-class classification head is replaced with a 2-class linear layer whose outputs correspond to `notpoop` and `poop`. In the current system, this classifier is not treated as the final event detector. Instead, it acts as a gate: if the MobileNetV2 poop-posture probability is high enough, the frame becomes a candidate event and can be evaluated by the CLIP confirmation stage.

Training setup. The posture dataset is loaded using PyTorch's `ImageFolder` format with explicit train, validation, and test splits. Training images are augmented using resized crops, horizontal flips, color jitter, and ImageNet normalization. These augmentations improve robustness to framing, dog orientation, camera distance, and outdoor lighting variation:

- `RandomResizedCrop(224)` encourages the classifier to rely on posture rather than exact image framing.
- `RandomHorizontalFlip` accounts for left-facing and right-facing dog poses.
- Mild `ColorJitter` simulates outdoor brightness, contrast, and camera white-balance changes.

Validation and test images use deterministic resizing, center cropping, and the same ImageNet normalization.

Optimization and threshold selection. Training uses cross-entropy loss with label smoothing ($\epsilon = 0.05$) to reduce overconfidence in borderline examples. AdamW is used as the optimizer, and cosine annealing gradually lowers the learning rate during training. A validation sweep over candidate thresholds is used to select the operating point $\tau^\star$ that maximizes F1. In the current results, the MobileNetV2 poop-posture classifier achieved 99.5% test accuracy and an F1 score of 0.994 at $\tau^\star = 0.53$. This threshold is used as the main reference point for runtime posture gating and hysteresis.

3.3 CLIP-Based Event Confirmation

Motivation. The original posture classifier was trained for poop/not-poop recognition, but the broader Nereus-LC problem requires distinguishing between pee, poop, and normal dog behavior. Training a reliable supervised classifier for all event types requires a larger and more balanced dataset than is currently available. To address this, the latest system uses CLIP as a lightweight confirmation layer before committing to a larger supervised pee/poop dataset.

Prompt groups. The CLIP stage scores the event crop against three prompt groups:

- **neutral:** normal dog behavior, such as standing, walking, sniffing, or sitting.
- **poop:** prompts describing a dog pooping, defecating, or squatting to poop.
- **pee:** prompts describing a dog peeing, urinating, lifting a leg, or squatting to pee.

The system computes a score for each class group and logs the strongest event label when the pee or poop score exceeds the neutral score by a sufficient margin.

Prompt sensitivity. Early testing showed that CLIP is sensitive to prompt design. Richer prompt sets can sometimes cause pee and poop labels to flip when the descriptions overlap too much. Minimal prompts can produce clearer separation in some clips, but they may also be less robust across different dogs, camera angles, and event types. This means CLIP is useful as a confirmation and logging tool, but it is not yet reliable enough to be the only event classifier.

CLIP score smoothing. Like the MobileNet posture score, the CLIP pee, poop, and neutral scores can fluctuate from frame to frame. The runtime therefore maintains an EMA-smoothed score for each CLIP class:

$$s_{t,c}^{\text{ema}} = \alpha s_{t,c} + (1-\alpha)s_{t-1,c}^{\text{ema}},$$

where $c \in \{\text{pee}, \text{poop}, \text{neutral}\}$. The smoothed CLIP scores are used to reduce single-frame label flicker and produce more stable event logs.

Event confirmation rule. A CLIP-confirmed event is logged when the strongest pee or poop score is high enough, exceeds the neutral score by a margin, and exceeds the competing event class by a margin. This prevents the system from labeling an event when CLIP is uncertain or when pee and poop scores are too close. In practice, these margins still need calibration using more real clips and hard-negative examples.

3.4 Event-Triggered Runtime Pipeline

Frame-level data flow. Each frame passes through *NereusWatch* as follows:

1. Read a frame from the camera or video file.
2. Detect the dog using YOLOv8.
3. Apply MobileNetV2 posture gating to the dog crop.
4. Smooth the posture score and apply ON/OFF hysteresis.
5. Send candidate event crops to CLIP for pee/poop/neutral scoring.
6. Log the event and save snapshots, crops, and annotated video.

3.5 Temporal Smoothing and Hysteresis

MobileNet EMA. Raw posture probabilities can fluctuate when the dog moves, becomes partially occluded, or transitions into or out of a squat posture. Without smoothing, the system could repeatedly cross the decision threshold and generate unstable event behavior. The MobileNet posture score is therefore smoothed using:

$$p_{\text{ema}} = \alpha p + (1-\alpha)p_{\text{ema,prev}}.$$

This acts as a temporal low-pass filter and requires the posture evidence to remain consistent across frames.

Figure 3: Example of raw posture probabilities versus the smoothed exponential moving average (EMA) used at runtime. EMA stabilizes transient fluctuations and prevents false event triggers.

On/off hysteresis. The runtime uses two thresholds instead of one:

$$\text{enter event if } p_{\text{ema}} \geq \tau_{\text{on}}, \qquad \text{clear event if } p_{\text{ema}} < \tau_{\text{off}}.$$

The ON threshold starts an event only when the smoothed score is high enough, while the lower OFF threshold prevents the system from immediately clearing the event because of one weak frame. This reduces flicker and makes the event decision more stable.

3.6 Target Region Cropping and Evidence Capture

Butt-region crop. When the posture gate remains active long enough, the system estimates the dog's heading from recent bounding-box center movement. It then locks a localized lower-body region near the likely waste location. This crop is not a segmentation result; it is a heuristic target region based on the dog bounding box, recent motion direction, and the expected location of the rear/lower-body area.

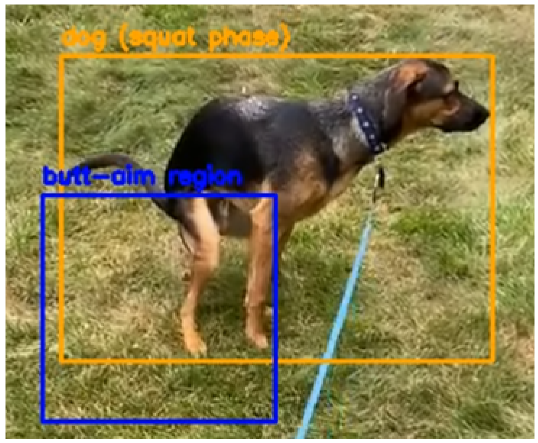


Figure 4: Automatically generated target-region crop. Once a sustained event posture is detected, *NereusWatch* locks a localized region for post-event inspection and review.

Snapshots and crops. The system saves multiple forms of evidence:

- clean reference snapshots when the scene is idle,
- post-event full-frame snapshots after the dog leaves,
- localized post-event region crops,
- matching clean crops when a prior clean frame is available,
- optional annotated MP4 video showing detections, scores, and target regions.

These outputs make the pipeline easier to evaluate and provide evidence for later physical-response integration.

3.7 State Machine

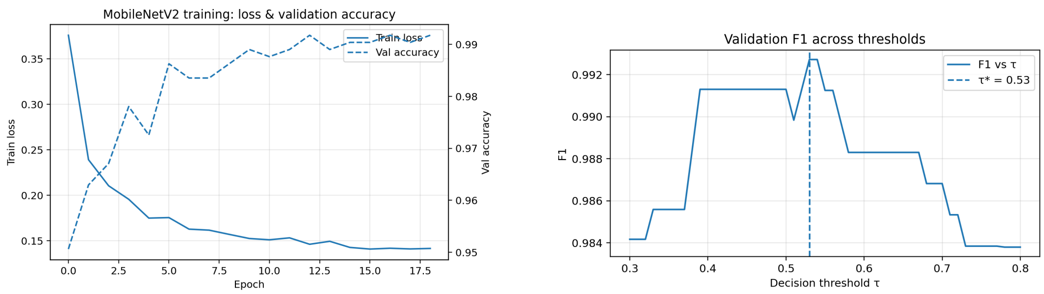
The finite-state machine governs high-level runtime behavior:

- **IDLE:** No dog is currently detected. The system may save clean reference snapshots when the scene is static.
- **DOG_SEEN:** A dog has been detected. YOLOv8 provides the bounding box, MobileNetV2 provides the posture-gate score, and CLIP can confirm likely pee/poop/neutral evidence from the event crop.
- **SQUAT:** A candidate waste-event posture has been held long enough to lock a target region. The system waits for the dog to leave before saving post-event evidence.
- **WAIT_CLEAR:** The system waits through a short cooldown so that temporary missed detections do not immediately reset the event.

3.8 Lessons from Earlier Pee-Detection Experiments

Earlier experiments tested supervised pee detection using YOLOv11 and ResNet18. These models showed that pee detection is harder than poop-posture detection because peeing can involve subtle rear-leg, tail, and body-angle cues. In one test, a ResNet18 model reached reasonable validation accuracy but predicted “pee” too aggressively on an unseen seven-second clip, suggesting overfitting and class bias. YOLOv11 experiments also showed overprediction on training videos. These results motivated the current CLIP-based confirmation layer and highlighted the need for more diverse positives, hard negatives, and calibrated thresholds.

4 Training Outputs



(a) Training loss and validation accuracy.

(b) Validation F1 across thresholds τ.

Figure 5: Training dynamics for the posture classifier.

5 Results

5.1 Experimental Setup

Hardware. Training and runtime experiments used an RTX-class GPU and a USB camera at native 640×480 resolution. YOLOv8 internally resizes frames to its 640-pixel inference size.

Metrics. Accuracy, precision, recall, and F1 are reported. A τ-sweep determines $\tau^\star$.

5.2 Posture Classifier Quantitative Results

Aggregate metrics. Test accuracy reached 0.995, with an F1 score of 0.994 at $\tau^\star = 0.53$.

Threshold selection. Figure 5b shows stable F1 performance between τ=0.45–0.60.

Confusion analysis. Figure 6 shows the confusion matrix. Errors mainly occur in transitional or partially occluded frames.

Precision–recall behavior. Figure 7 shows an average precision near 1.0 and strong separation between classes.

5.3 Runtime Pipeline Behavior

Qualitative behavior. In qualitative runtime tests, the pipeline successfully detected dogs, displayed MobileNet and CLIP scores, logged CLIP-confirmed event labels, and saved post-event snapshots and localized crops. EMA smoothing and hysteresis reduced frame-to-frame flicker, but event quality still depended on camera angle, threshold settings, and whether CLIP could clearly separate pee, poop, and neutral prompts.

5.4 CLIP Prompt Bias and Sensitivity Results

Minimal prompt behavior. CLIP was tested as a pee/poop/neutral confirmation layer using multiple prompt sets on videos that primarily contained dogs pooping or performing normal non-event behavior. With a minimal prompt set, CLIP often assigned the highest score to the pee class even when the dog was walking, sniffing, or in a poop-related event. In one test, pee scores stayed around

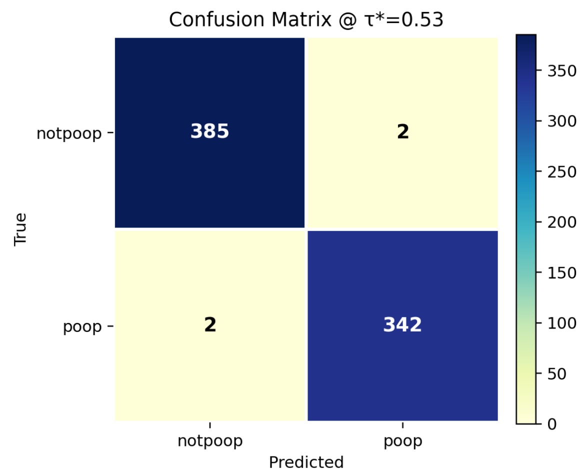


Figure 6: Confusion matrix at $\tau^\star = 0.53$.

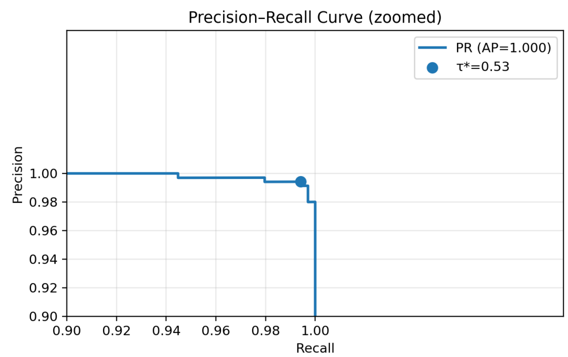


Figure 7: Precision–recall curve at $\tau^\star = 0.53$.

0.536–0.653, poop scores stayed around 0.406–0.421, and neutral scores stayed around 0.028–0.031. This showed that the minimal prompt set did not reliably separate pee, poop, and neutral behavior; instead, it introduced a strong pee bias.

Richer prompt behavior. A richer prompt set was also tested to evaluate whether more descriptive text improved CLIP confirmation. The neutral group included common non-event behaviors such as standing, walking, sniffing, and sitting. The poop group included descriptions such as pooping, defecating, and squatting to poop, while the pee group included peeing, urinating, leg-lift, and squatting-to-pee descriptions. This richer prompt set reduced the absolute pee score, but pee and poop still remained close: pee scores were approximately 0.190–0.259, poop scores were approximately 0.134–0.191, and neutral scores remained low at approximately 0.023–0.024.

Interpretation. These results show that CLIP prompt wording can strongly bias event labels. In this experiment, CLIP tended to favor pee even when the input video showed poop-related or neutral dog behavior. Adding richer prompts changed the score scale but did not fully solve the class-separation problem. The main lesson is that CLIP cannot be treated as a drop-in classifier without careful prompt engineering, hard-negative examples, and calibrated thresholds. For NEREUS-LC, CLIP is useful for exploratory confirmation and logging, but it is not yet reliable enough to drive an unattended physical response.

5.5 Lessons Learned

Prompt engineering matters more than expected. The CLIP experiments showed that prompt wording is a real design variable, not just a formatting detail. CLIP compares image crops against the supplied text descriptions, so small wording changes can shift the relative pee, poop, and neutral scores. This made prompt engineering one of the main empirical lessons from the project.

The problem shifted from classification to event reliability. The MobileNetV2 poop-posture classifier performed very well on held-out images, but runtime video introduced a different problem. A real event unfolds over time, and each frame may include blur, partial occlusion, changing posture, or unclear evidence. As development progressed, the main challenge became less about classifying isolated images and more about making stable event-level decisions. In the current pipeline, YOLOv8 localizes the dog, MobileNetV2 gates likely candidate behavior, CLIP provides event confirmation, and temporal smoothing converts noisy frame-level scores into more stable decisions.

5.6 Limitations

The current system is still limited by the size and diversity of the available event data. Pee, poop, and neutral dog behaviors can look visually similar, especially when the dog is partially occluded, viewed from an unfavorable angle, or captured during a transition pose. CLIP is also sensitive to prompt wording, so its scores should be interpreted as confirmation evidence rather than ground truth. In the current experiments, CLIP showed a strong tendency to favor the pee label, even for poop-related or neutral behavior. Additional real outdoor clips, hard-negative examples, prompt tuning, and calibrated event thresholds are needed before the system can be considered reliable for unattended physical response.

6 Future Work After This Course

6.1 Short-Term Extensions

The next step is to improve event-level reliability. This includes collecting more real outdoor clips, adding hard-negative examples such as sniffing, sitting, stretching, and non-event squats, and tuning CLIP prompt groups so pee, poop, and neutral labels are better separated. The CLIP confirmation thresholds should also be calibrated using a larger validation set instead of hand-selected margins.

6.2 Medium-Term System Development

The runtime pipeline should be connected to a physical response mechanism, such as a sprinkler, deterrent cue, or other localized station action. Additional work is also needed to evaluate when the system should respond immediately, when it should only log evidence, and how to avoid unnecessary responses to uncertain detections.

6.3 Long-Term Research Directions

Longer-term work may revisit ExG/VARI lawn-health analysis, IR sensing, or a mobile platform, but these are no longer the core implementation of the current prototype. The main research direction is to determine whether a hybrid pipeline using object detection, posture gating, vision-language confirmation, and temporal smoothing can produce reliable event-level decisions in real outdoor settings.

7 Conclusion

NEREUS-LC represents an early step toward event-triggered, vision-based dog waste detection and response. The current system combines YOLOv8 dog detection, MobileNetV2 posture gating, CLIP-based pee/poop/neutral confirmation, EMA smoothing, hysteresis, target-region cropping, and event logging. The MobileNetV2 poop-posture classifier achieved strong standalone performance (0.995 test accuracy, $F1 = 0.994$ at $\tau^\star = 0.53$), but the broader runtime problem requires more than still-image classification.

The main empirical lesson was that CLIP prompt wording strongly affects event labels. In the current tests, CLIP often favored the pee label even for poop-related or neutral behavior, showing that prompt engineering, hard negatives, and calibrated thresholds are necessary before using CLIP for unattended response.

Overall, the current prototype demonstrates a practical hybrid test harness for moving from frame-level model predictions toward stable event-level behavior. Continued development will connect the perception pipeline to a physical response mechanism and evaluate whether the system can reliably support targeted lawn-care intervention in real-world conditions.

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